EDGE DEVICE MEASUREMENT REPORT

Electronic Nose Coffee Classification System

Platform: NVIDIA Jetson Orin Nano

Model: Random Forest

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MEASUREMENT SUMMARY

Single-Sample Inference Latency	Avg Total Power (Training)	Training Time
0.001079 ms	**7.33 W**	**20.68 s**

MODEL CONFIGURATION

Model Type	Random Forest
Number of Trees	50
Extracted Features	mean, std, min, max, range, median
Feature Dimensions	6 statistics x 8 sensors = 48 features
Single-Sample Inference Latency	0.001079 ms
Model Size	71.48 KB
Number of Parameters	1500
Memory Footprint	3975.00 MB
Power Measurement	7.33 W

POWER, THERMAL & RESOURCE MEASUREMENTS DURING TRAINING

Power Metric	Value	Thermal / Energy	Value
Avg Total Power	7.33 W	**Avg CPU Temp**	52.2 C
Peak Total Power	9.27 W	**Peak CPU Temp**	53.4 C
Avg CPU+GPU Power	2.42 W	**Avg GPU Temp**	52.2 C
Peak CPU+GPU Power	3.85 W	**Peak GPU Temp**	52.7 C
Avg RAM Usage	3711 MB	**Avg GPU Util**	30.7 %
Peak RAM Usage	3975 MB	**Peak GPU Util**	93.0 %
Energy Used	145.2 J (0.0403 Wh)	**Training Duration**	20.68 s

GAS SENSORS USED (8 Sensors)

TGS2600	**MQ135**	**MQ3**	**MQ6**
MQ7	**TGS2602**	**TGS2611**	**TGS2620**